

Causal Prefix–Sinkhorn Attention: Scale-Stable Balancing and Role-Complete Routing

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Abstract. Column balancing is attractive in attention because ordinary row-softmax leaves key usage unconstrained, yet a naive Sinkhorn iteration over a causally masked token-attention matrix is not causal: future query rows enter past column normalizers. Causal Prefix–Sinkhorn replaces global column sums by cumulative prefix masses, giving an exact zero-future-gradient guarantee and a streaming scan implementation. This revision completes the operator’s theory and its first falsification program. First, one exact prefix iteration has a forced-diagonal mode: the current token is the unique row maximum. Second, the only nonnegative lower-triangular matrix with unit row and column sums is the identity, so causal balancing is correctly understood as a finite routing regularizer rather than a convergent causal projection onto a rich bistochastic class. A log-domain formulation removes float32 underflow; a residual-aware diagonal debit has an exact virtual-exposure interpretation; and an explicit reservoir produces a causal substochastic token block. Most importantly, the scale falsifier is decisively passed: the Softmax-minus-Prefix gain rises from 0.984 bpb at width 128 to 1.182 at width 512 and remains 1.184 at width 1024. Pure local or balanced attention, however, fails adversarial lag-48 copy. We therefore introduce role-complete causal attention, assigning heads to local, balanced, and untouched free-softmax routes. A single free head restores lag-48 accuracy to 1.000 in every tested route program. On variable-lag copy trained uniformly over lags 8–48, $2L + 1B + 1F \rightarrow F$ reaches 0.9293 mean accuracy on unseen lags 56–80, ahead of Recency $\rightarrow F$ at 0.9220, RAPS $\rightarrow F$ at 0.8832, Softmax at 0.7745, and $3L + 1F \rightarrow F$ at 0.4791. On a sealed three-seed byte-corpus confirmation, $3L + 1F$ reaches 4.7203 ± 0.0154 bpb and $2L + 1B + 1F$ reaches 4.7268 ± 0.0100 , improving over Softmax by 0.0997 and 0.0932 bpb with 3/3 seed wins. Placing the specialist layer below a pure-softmax readout improves the gain to 0.1019 bpb. The resulting view is architectural: Prefix–Sinkhorn is a novel, scale-stable causal balancing primitive, and its strongest use is as one route inside a system that preserves an explicit unrestricted path.

1 Introduction

Softmax attention normalizes each query row independently. It does not constrain the columns: a small set of keys may absorb mass from many queries, and deep pure-attention stacks exhibit a strong rank-one bias [4, 6, 11]. Doubly stochastic attention addresses the column side by alternating row and column normalizations, with demonstrated optimization and rank-preservation benefits in bidirectional Transformers [2, 7, 8, 9, 10]. The autoregressive case is not obtained by merely placing a triangular mask inside the same iteration. Column normalization couples query rows, including future rows, and therefore creates a hidden future channel.

Causal Prefix–Sinkhorn attention was introduced to close that channel. Starting from ordinary causal row-softmax, it replaces each global key-column sum by the cumulative mass visible at the current query row. The operation is a prefix scan, is strictly causal by construction, and retains the same $O(T^2)$ attention complexity. The first working draft established three headline facts: the masked full-support adaptation reached an impossible 0.156 bpb because it leaked; the prefix construction had exactly zero future gradient; and under an identical small-scale TinyStories budget it improved validation loss from 3.284 to 2.348 bpb, a gain of 0.936 bpb.

Version 0.5 resolves the questions that remained open after that first result. The primary falsifier was explicit: if the gain vanished as width increased, Prefix–Sinkhorn would be a small-model artifact. It does the opposite. The $k = 1$ gain increases across widths 128, 256, 384, and 512, then stabilizes at 1.184 bpb at width 1024. The result is not an optimizer curiosity that capacity washes away; it is a scale-stable routing effect.

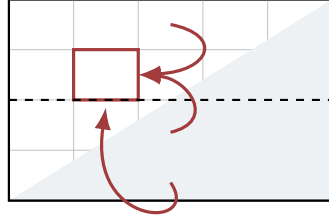
The new theory also changes the right architectural interpretation. Exact causal double stochasticity has only one lower-triangular solution, the identity. A single Prefix–Sinkhorn iteration already makes the diagonal the unique maximum of each row. These are not defects. They identify the operator

precisely: it is an intentionally finite pressure toward balanced, local, non-hoarding flow. Applying that pressure to every head removes an unrestricted long-range path and fails adversarial delayed copy. The solution is *role completion*: local heads encode a strong recency prior, balanced heads perform causal prefix allocation control, and at least one free head remains ordinary causal softmax.

The contributions are:

- a strict causal construction and a standard zero-future-gradient acceptance test for nonstandard normalizers;
- two exact structural results—forced diagonal mode and the lower-triangular bistochastic obstruction—that characterize the operator’s geometry;
- numerically stable Log-Prefix, residual-aware Prefix–Sinkhorn (RAPS), and a causal reservoir extension;
- a completed scale campaign through width 1024, showing a stable gain of about 1.18 bpb rather than decay;
- role-complete causal attention with an explicit free-head invariant, 1.000 lag-48 copy, and three-seed sealed-test gains;
- variable-lag interpolation and extrapolation through lag 80, where the balanced role-complete route is the strongest tested long-range program;
- a depth-routing result: specialize the foundation layer and keep the upper readout layer free;
- a reproducible implementation with 69 automated tests covering causality, support, streaming equivalence, route assignment, and training behavior.

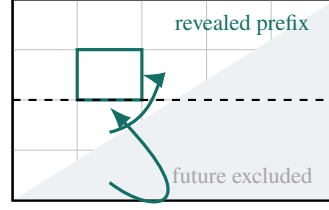
(a) Masked full-support balancing



$$C_j = \sum_{r=1}^T A_{rj}$$

future rows enter the denominator

(b) Causal prefix balancing



$$S_{ij} = \sum_{r=1}^i A_{rj}$$

normalizer closes at row i

Figure 1: Triangular support is not computational causality. Full-support column normalization feeds future query rows into a past denominator. Prefix–Sinkhorn closes each column mass at the current row, so the normalizer depends only on the revealed prefix.

2 From masked Sinkhorn to a causal scan

2.1 The leak in the naive adaptation

Let $\mathbf{L} \in \mathbb{R}^{T \times T}$ be token-attention logits and let $\mathbf{M}_{ij} = 0$ for $j \leq i$ and $-\infty$ otherwise. Ordinary causal attention is

$$\mathbf{A}^{(0)} = \text{softmax}(\mathbf{L} + \mathbf{M}) \quad \text{row-wise.} \quad (1)$$

A naive masked Sinkhorn step normalizes the columns over the full triangular matrix and then re-normalizes the rows:

$$\tilde{A}_{ij} = \frac{A_{ij}}{\sum_{r=1}^T A_{rj}}, \quad A'_{ij} = \frac{\mathbb{1}[j \leq i] \tilde{A}_{ij}}{\sum_{m \leq i} \tilde{A}_{im}}. \quad (2)$$

The denominator of A'_{ij} includes A_{rj} for $r > i$. The matrix is triangular, but the computation is not causal. In the original A/B, this operator reached 0.156 bpb and a future-input gradient norm of 0.385. The impossible loss was the leak signal, not a language-modeling result.

Acceptance probe. Choose a position i , backpropagate a loss that depends only on the output at i , and inspect

$$g_{\text{future}} = \left\| \frac{\partial \ell_i}{\partial x_{j>i}} \right\|_2. \quad (3)$$

A causal normalizer must give exact zero up to machine precision. This probe is cheap enough to be a unit test and catches channels that a loss curve does not.

2.2 Causal Prefix–Sinkhorn

Replace the global column mass in eq. (2) by a cumulative prefix mass. For iteration $t = 1, \dots, k$,

$$S_{ij}^{(t)} = \sum_{r=1}^i A_{rj}^{(t-1)}, \quad (4)$$

$$\tilde{A}_{ij}^{(t)} = \frac{A_{ij}^{(t-1)}}{S_{ij}^{(t)} + \varepsilon}, \quad (5)$$

$$A_{ij}^{(t)} = \frac{\mathbb{1}[j \leq i] \tilde{A}_{ij}^{(t)}}{\sum_{m \leq i} \tilde{A}_{im}^{(t)} + \varepsilon}. \quad (6)$$

The output remains $\mathbf{A}^{(k)}V$. One iteration adds a cumulative sum over query rows, elementwise division, and row normalization. The cumulative sum is a parallel prefix scan.

Proposition 1 (Strict causality). *For every k , row i of eq. (6) is a function only of rows $1, \dots, i$ of $\mathbf{A}^{(0)}$. Therefore the layer output at i depends only on tokens $1, \dots, i$, and $\partial y_i / \partial x_j = 0$ for all $j > i$.*

Proof. $\mathbf{A}^{(0)}$ is causal by masked row-softmax. The prefix mass S_{ij} uses rows $r \leq i$ only. Elementwise scaling preserves that dependency set, the mask is constant, and row normalization uses only row i . Induction over the iteration count completes the proof. Multiplication by V reads v_j only for $j \leq i$ because the final matrix remains lower triangular. $\square$

The numerical probe gives future-gradient norm exactly zero for Softmax, Log-Prefix, RAPS, the reservoir extension, and all fixed route programs; the intentionally leaky full-support control gives 0.50446 in the current audit.

3 Exact geometry of causal balancing

The original interpretation described Prefix–Sinkhorn as an online approximation to doubly stochastic balancing. That remains operationally useful, but the causal triangular geometry has two exact consequences that determine the correct endpoint.

3.1 One step forces the diagonal mode

Assume $\mathbf{A}$ is a positive lower-triangular row-stochastic matrix, as produced by masked softmax on finite logits, and take one exact Prefix–Sinkhorn step with $\varepsilon = 0$.

Theorem 1 (Forced-diagonal mode). *For every row i , the pre-renormalized diagonal entry is exactly one and every earlier-key entry is strictly smaller than one:*

$$\frac{A_{ii}}{S_{ii}} = 1, \quad \frac{A_{ij}}{S_{ij}} < 1 \quad (j < i). \quad (7)$$

Consequently $A'_{ii} > A'_{ij}$ for every $j < i$: the diagonal is the unique maximum of row i after one exact prefix iteration.

Proof. Because column i first becomes available at row i , $S_{ii} = A_{ii}$. For $j < i$, the prefix sum contains both A_{ij} and the strictly positive diagonal term A_{jj} , hence $S_{ij} > A_{ij}$. Row normalization preserves the strict ordering. $\square$

The theorem was audited on 4,096 random matrices comprising 173,950 rows with zero float64 violations. A direct probability-domain float32 implementation produced nine violations under extreme underflow; the log-domain implementation in section 4.1 produced zero.

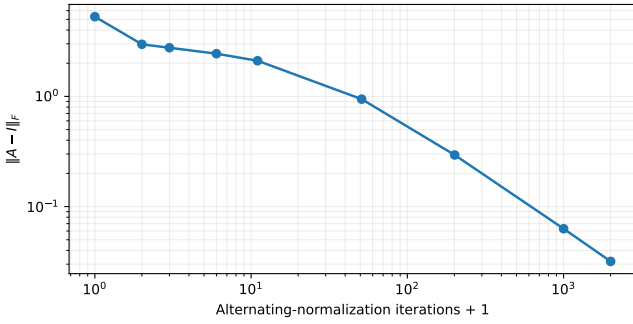


Figure 2: Repeated triangular alternating normalization approaches the unique bistochastic boundary point, the identity. Prefix–Sinkhorn is useful as a finite routing pressure, not as a converged causal projection.

3.2 The causal bistochastic obstruction

Theorem 2 (Lower-triangular bistochastic obstruction). *If $A \geq 0$ is lower triangular and every row and column sums to one, then $A = I$.*

Proof. Column T contains only A_{TT} , so its unit sum forces $A_{TT} = 1$. The unit sum of row T then forces all other entries in that row to zero. Remove the final row and column and repeat by backward induction. $\square$

Corollary 1. *There is no rich family of exact, strictly causal, doubly stochastic token-attention matrices. Repeated exact balancing over a lower-triangular support converges toward the identity boundary point.*

The numerical alternating-normalization audit follows this prediction: the Frobenius distance to identity falls from 5.290 before iteration to 0.0318 after 2,000 iterations (fig. 2). The practical operator is therefore the finite step—and the iteration sweep confirms it. At width 192, $k = 1$ reaches 2.1784 bpb, $k = 2$ reaches 2.1854, and $k = 5$ degrades to 2.4340. One step is the canonical operator.

4 Stable and residual-aware operators

4.1 Log-Prefix

Probability-domain cumulative masses can underflow when masked logits span hundreds of nats. The stable formulation stays in log space. Let

$$\ell_{ij} = \log \text{softmax}(\mathbf{L} + \mathbf{M})_{ij}, \quad (8)$$

and compute

$$u_{ij} = \text{logcumsumexp}_{r \leq i} \ell_{rj}, \quad (9)$$

$$r_{ij} = \ell_{ij} - \alpha u_{ij}, \quad (10)$$

$$A'_{i:} = \text{softmax}(r_{i:}). \quad (11)$$

The original operator is $\alpha = 1$. $\alpha = 0$ recovers Softmax and intermediate values continuously control prefix pressure. The implementation uses $\varepsilon = 0$ by default and preserves the exact causal support. Batch and streaming implementations agree within 1.192×10^{-7} in float32.

4.2 Residual-Aware Prefix–Sinkhorn

The forced-diagonal theorem identifies a useful control knob. RAPS applies a diagonal debit $\delta \geq 0$ before the final row softmax:

$$r_{ii} \leftarrow r_{ii} - \delta. \quad (12)$$

This has an exact exposure interpretation. Multiplying the diagonal raw weight by $e^{-\delta}$ is equivalent to adding virtual historical exposure

$$E_{ii}^{\text{virtual}} = (e^{\delta} - 1)A_{ii}. \quad (13)$$

For the canonical $\delta = 3$, the multiplier is 19.085537; the two constructions agree to 4.441×10^{-16} . RAPS retains causal prefix budgeting while preventing the diagonal mode from consuming an entire head.

4.3 Causal prefix reservoir

A second extension adds an explicit null route with logit ρ . The token block then satisfies

$$\sum_{j \leq i} A_{ij} \leq 1, \quad (14)$$

with the remaining mass allocated to the reservoir and carried by the residual stream. This escapes the identity obstruction by moving from a stochastic triangular token block to a sub-stochastic one. In the random audit, mean reservoir mass is 0.0351 with range 0.0015–0.6660. The extension is strictly causal and streaming-equivalent within 1.192×10^{-7} .

5 Role-complete causal attention

The theory predicts a sharp specialization tradeoff. Prefix balancing and strong recency are excellent local allocation priors; neither should be required to carry all long-range retrieval. We therefore assign exact roles to heads.

5.1 Three routes

For head h and causal logits $L^{(h)}$:

Local route L. A strictly-past recency head,

$$A_{ij}^{(h)} \propto \mathbb{I}[j < i] \exp(L_{ij}^{(h)} - \beta(i - j)), \quad (15)$$

with row 1 mapped to itself. The confirmed route programs use $\beta = 0.8$.

Balanced route B. Log-Prefix/RAPS with $\alpha = 1$ and diagonal debit $\delta = 3$.

Free route F. Untouched causal Softmax,

$$A^{(h)} = \text{softmax}(L^{(h)} + M). \quad (16)$$

A role program is denoted $n_L \mathbf{L} + n_B \mathbf{B} + n_F \mathbf{F}$. The architectural invariant is

$$n_F \geq 1. \quad (17)$$

This preserves an entire unrestricted value subspace, not merely a fractional mixture. The residual stream already supplies an identity path; the screens show that a dedicated self head is unnecessary in the strongest configurations.

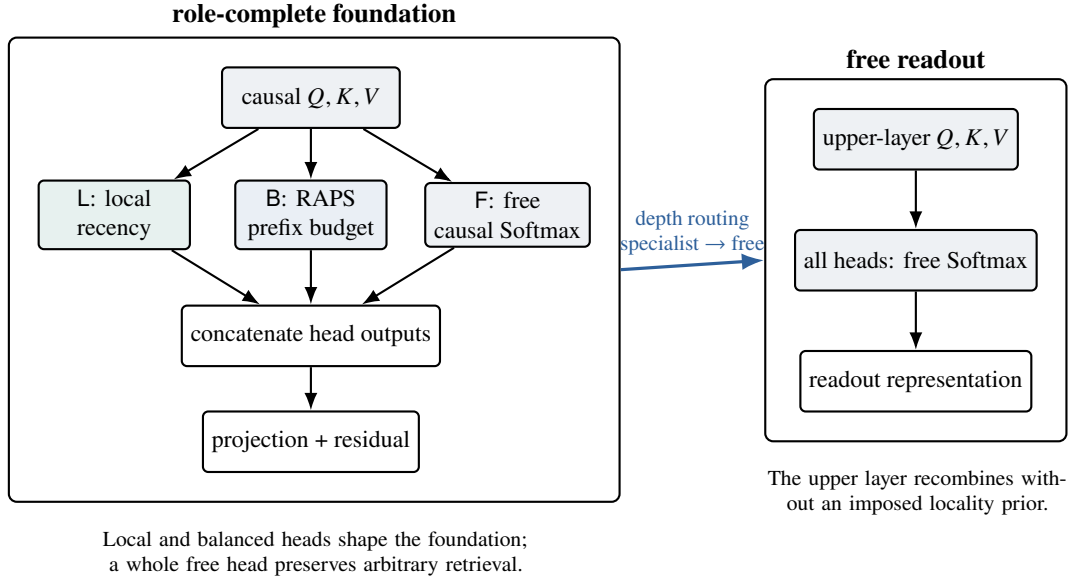


Figure 3: Role-complete causal attention and the best confirmed depth pattern. Head roles are structural. A whole free head is an invariant and restores adversarial long-range copy. Across depth, specialization works best below a pure-Softmax readout layer.

5.2 Why a whole free head matters

A per-query adaptive mixture with a 0.25 free probability floor still failed lag-48 copy at 0.038 accuracy: fractional access to the free branch did not preserve a complete long-range value subspace. Replacing the floor by one untouched free head solved the problem immediately. Every one of eleven tested role programs satisfying eq. (17) reached 1.000 lag-48 copy. Pure RAPS reached 0.0647; pure recency and the original fully specialized dual route reached 0.0152.

5.3 Depth routing

The 24-program depth screen crossed local, balanced, role-complete, and free layers in both orders. The best validation program was $2L + 1B + 1F \rightarrow F$; the second was $3L + 1F \rightarrow F$. In the three-seed sealed-test confirmation, the latter reached 4.7181 ± 0.0173 bpb, a paired gain of 0.1019 ± 0.0150 over $F \rightarrow F$, with 3/3 seed wins. Reversing the order is materially weaker: free foundation layers followed by specialist readouts sit near the bottom of the validation screen. Local allocation belongs in the foundation; unrestricted recombination belongs above it.

6 Experiments

6.1 Protocols

The paper uses three distinct protocols and keeps their evidential roles separate.

TinyStories byte modeling. The original A/B uses a decoder-only Transformer with four blocks, width 192, four heads, context 256, learned positional embeddings, AdamW at 3×10^{-4} , weight decay 0.01, gradient clipping 1.0, batch 24, and $k = 1$. The scale campaign keeps the identical 400-step comparison protocol and extends width through 1024.

Table 1: Original identical-budget TinyStories A/B and direct leak probe. Lower bpb is better.

Attention	Causal	bpb @ 400	Future grad.
Softmax	yes	3.284	0
Masked full-support	no	0.156 [†]	0.385
Prefix–Sinkhorn, $k = 1$	yes	2.348	0

[†]Invalid: future leakage.

Sealed archive byte corpus. A deterministic file manifest partitions a pre-existing research archive into byte-level train, validation, and test files. The role screens use validation only. The file-hash test split is opened only for the named three-seed confirmation stages. Models have width 48, four heads, two layers, context 96, batch 12 or 16, and 260 training steps; fixed-route models contain 85,440 parameters. This corpus is a mechanism confirmation set, not a replacement for TinyStories.

Synthetic long-range stress. Fixed delayed copy uses lag 48 in context 96. The loss is applied only after the lag becomes available. The test is intentionally adversarial to local and prefix-budget routes: success requires preserving a free distant path. A second task samples the training lag uniformly from 8 through 48 for 900 steps and evaluates fixed lags 8, 16, ..., 80; lags 56–80 are strict extrapolation beyond the training support. Each variable-lag arm is run on seeds 8317 and 9137. Associative-recall generators are included in the release, but current low Softmax ceiling prevents using them as a headline discriminator; they remain a calibration target.

All fixed operators use exact causal masks. Confirmation seeds are 1908, 2408, and 42017. Reported uncertainties are across these three runs. Test files remain unopened during selection.

6.2 Original construction and leak falsification

Prefix–Sinkhorn improves the valid baseline by 0.936 bpb and remains stable over the 100-times longer horizon reported in

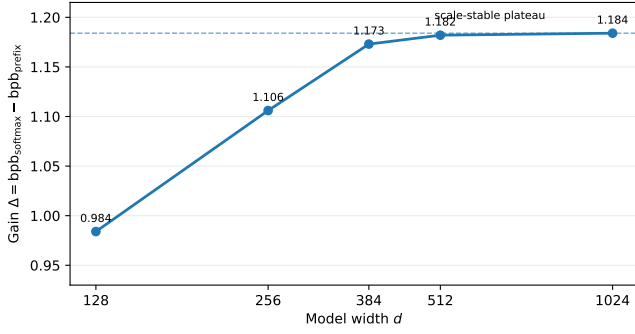


Figure 4: Completed $k = 1$ width sweep. The Prefix–Sinkhorn gain grows through width 512 and remains 1.184 bpb at width 1024.

Table 2: Audit summary. All valid operators have exact causal support.

Audit item	Result
Forced-diagonal random matrices	4,096
Rows checked / violations (float64)	173,950 / 0
Float32 violations: probability / log	9 / 0
Max batch–stream error	1.192×10^{-7}
RAPS exposure identity error	4.441×10^{-16}
Future grad., valid operators	0
Future grad., leaky control	0.50446
Automated test suite	69 / 69 pass

the original draft: 1.100 bpb after 76 million bytes with no normalizer instability.

6.3 The scale falsifier is passed

The completed $k = 1$ series is

$$\begin{array}{c|ccccc} d & 128 & 256 & 384 & 512 & 1024 \\ \hline \Delta \text{bpb} & 0.984 & 1.106 & 1.173 & 1.182 & 1.184 \end{array} \quad (18)$$

where $\Delta \text{bpb} = \text{bpb}_{\text{softmax}} - \text{bpb}_{\text{prefix}}$. The gain increases by 20.1% from width 128 to 512 and then changes by only 0.002 bpb at width 1024. The falsifier “gain vanishes with scale” is not approached; the curve reaches a scale-stable plateau near 1.18 bpb (fig. 4). The associated column-deviation measurement is 0.105 for Prefix–Sinkhorn versus 0.899 for Softmax, an 8.6-times movement toward balanced column usage.

6.4 Operator and numerical audit

The log-domain form is behaviorally identical to the probability-domain form away from underflow, while remaining finite in extreme cases. The streaming equality covers Log-Prefix, reservoir Prefix, dual route, and the role-separated self/local/balanced construction.

6.5 Three-seed sealed-test confirmation

The best raw score belongs to $3L + 1F$. The balanced route remains a strong, distinct primitive: $2L + 1B + 1F$ retains explicit prefix budgeting, wins 3/3 seeds, and gives up only 0.0065 bpb to the top route. In the earlier fully specialized comparison, replacing the fourth recency head by a Prefix/RAPS head improved sealed test from 4.7314 to 4.7245 bpb, also on three

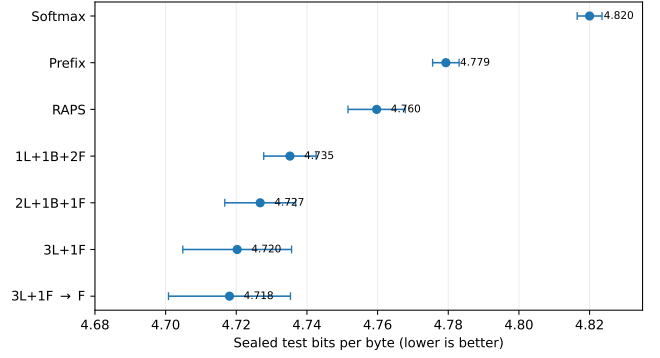


Figure 5: Sealed three-seed test bpb for the main operator and route progression. The depth-routed $3L + 1F \rightarrow F$ program is best.

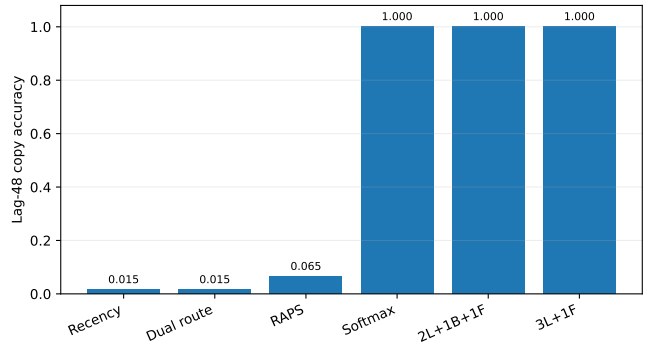


Figure 6: Lag-48 copy in context 96. Pure local or balanced programs collapse; every tested role-complete program with a whole free head reaches 1.000.

seeds. Prefix balancing therefore contributes beyond recency, but its correct role is specialized rather than universal.

6.6 Long-range preservation

The long-range result is categorical. Softmax reaches 1.000. RAPS reaches 0.0647; pure recency and the original dual route reach 0.0152. Every tested role-complete program with at least one untouched free head reaches 1.000, including all distributions of local and balanced roles. This is the free-head invariant in direct experimental form.

6.7 Variable-lag interpolation and extrapolation

The fixed-lag test establishes the existence of an unrestricted path; the variable-lag test asks whether the architecture can use that path across a continuum of distances. Training samples lags uniformly from 8 through 48. Evaluation extends to lag 80 without any further optimization. Results are means over two independent seeds (fig. 7).

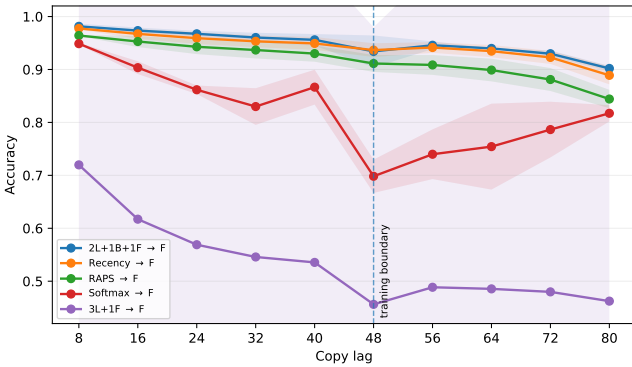
The decisive comparison is between the two role-complete foundations. $3L + 1F \rightarrow F$ is the best current byte-model route, yet reaches only 0.4791 on unseen lags. Replacing one local head by a balanced RAPS head raises unseen accuracy to 0.9293, the best result in the screen and 0.1548 above Softmax. At lag 80 it retains 0.9020 accuracy, versus 0.8173 for Softmax and 0.4624 for the pure-local route. The balanced head therefore carries a distinct allocation role that is not reducible to a static recency prior: Recency $\rightarrow F$ is strong at

Table 3: Three-seed sealed-test confirmation on the archive byte corpus. Negative Δ is better than Softmax. Every role-complete route wins on every seed.

Operator	Validation bpb	Test bpb	Paired Δ vs Softmax	Seed wins	Tokens/s
3L + 1F	4.5272 ± 0.0213	4.7203 ± 0.0154	-0.0997 ± 0.0128	3/3	$34,256 \pm 12,327$
2L + 1B + 1F	4.5290 ± 0.0201	4.7268 ± 0.0100	-0.0932 ± 0.0073	3/3	$15,497 \pm 5,572$
1L + 1B + 2F	4.5368 ± 0.0167	4.7352 ± 0.0074	-0.0848 ± 0.0040	3/3	$17,182 \pm 7,913$
Softmax	4.6067 ± 0.0215	4.8200 ± 0.0035	0	0/3	$37,084 \pm 9,562$

Table 4: Variable-lag copy. “Unseen” averages lags 56, 64, 72, and 80.

Foundation $\rightarrow$ readout	Train-range	Unseen
2L + 1B + 1F $\rightarrow$ F	0.9622	0.9293
Recency $\rightarrow$ F	0.9572	0.9220
RAPS $\rightarrow$ F	0.9397	0.8832
F $\rightarrow$ F	0.8514	0.7745
3L + 1F $\rightarrow$ F	0.5739	0.4791

**Figure 7:** Variable-lag copy accuracy. The dashed line marks the largest training lag. The balanced role-complete foundation is strongest on both interpolation and extrapolation; the pure-local route collapses despite its excellent byte-model score. Shading is ± 1 standard deviation across two seeds.

0.9220, but loses to the balanced route in aggregate and at the furthest lag.

6.8 Depth routing and bounded adaptivity

The three-seed depth confirmation is shown in table 5. The specialist-foundation/free-readout order dominates the reverse order in the 24-program screen.

Longer horizon and architecture scaling. The same archive-corpus program was extended from 260 to 1200 optimization steps with three seeds and the test split still unopened. The advantage grows rather than disappearing: 3L+1F $\rightarrow$ F reaches 3.7996 ± 0.0528 validation bpb, a paired gain of 0.2507 over F $\rightarrow$ F, while 2L + 1B + 1F $\rightarrow$ F reaches 3.8115 ± 0.0636 , a gain of 0.2388. A separate two-seed architecture-scale sweep shows that the canonical Prefix-preserving program strengthens with width: paired gains are 0.0333, 0.1757, and 0.2237 bpb at widths 32, 64, and 96, with 2/2 wins at every width. At depths 1, 3, and 4 the gains are 0.1191, 0.1352, and 0.1261 bpb, again 2/2 at every depth. Thus role completion is neither a short-run transient nor a shallow-stack artifact.

Table 5: Three-seed depth routing. Arrow order is bottom $\rightarrow$ top.

Program	Test bpb	Gain
3L + 1F $\rightarrow$ F	4.7181	0.1019
2L + 1B + 1F $\rightarrow$ F	4.7260	0.0940
Self/local/balanced $\rightarrow$ F	4.7324	0.0876
F $\rightarrow$ F	4.8200	0

A fully adaptive query gate was not competitive and destroyed copy despite a free-probability floor. Whole-head specialists recovered copy but mostly converged back to fixed local roles. The strongest current adaptive construction therefore keeps the proven structure as an anchor and permits only a bounded residual route budget:

$$g_{h,i} = (1 - \eta)e_{r_h} + \eta \text{softmax}(z_{h,i}), \quad (19)$$

while retaining one whole free head and a pure-softmax top layer. In the first validation screen, $\eta = 0.5$ reaches 4.6093 bpb versus 4.6107 for its fixed 3L + 1F anchor and 4.6799 for Softmax. This is an exploratory single-seed result, not used in the headline claims; it defines the next confirmation target.

7 Mechanism and interpretation

7.1 Prefix pressure is not merely a diagonal bias

The forced-diagonal theorem explains why Prefix–Sinkhorn can create a strong self path, but a matched diagonal-bias Softmax control does not reproduce the gain in the mechanism suite. Likewise, stopping gradients through the cumulative denominator preserves the same forward matrix but retains only about one quarter of the gain. The operator acts through both its forward allocation geometry and the backward coupling induced by prefix usage. The causal denominator makes each row compete with the already revealed allocation history of every key.

7.2 Recency is strong, balance is distinct

A linear recency prior explains a large part of the byte-model improvement and is the strongest single fixed route in several small screens. That fact improves the architecture: local heads should be explicit instead of forcing Prefix–Sinkhorn to impersonate all locality. The balanced route remains distinct in four ways. It measures actual prefix allocation rather than age; it reduces column concentration directly; in the fully specialized three-seed comparison, three local heads plus one Prefix/RAPS head beat four recency-like heads; and on variable-lag extrapolation the 2L + 1B + 1F $\rightarrow$ F program reaches 0.9293 versus 0.9220 for Recency $\rightarrow$ F, while the

pure-local role-complete program collapses to 0.4791. Role separation converts a mechanism ambiguity into a design: give recency and prefix balance different heads.

7.3 The free route is a guarantee, not a learned preference

The failed floor-gated adaptive model is decisive. A soft mixture can assign nonzero free mass and still destroy the long-range channel because all value dimensions are blended through the same constrained heads. A whole free head preserves an exact subspace on which attention can place arbitrary causal mass. This is stronger than a regularizer, a floor, or an initialization. It is a structural guarantee.

7.4 Why specialization belongs below the read-out

The depth screen shows an asymmetric composition law. A specialized bottom layer improves local feature formation and usage allocation; an unrestricted top layer can then recombine those features across the entire prefix. Reversing the order asks the final layer to discard access that the lower free layer preserved. The measured program order therefore matches the computational role of depth: structure first, unrestricted readout second.

8 Positioning and novelty

Sinkhorn balancing itself is classical [1]. Sinkformers applies doubly stochastic normalization to bidirectional token attention [2]; ESPFormer, LOTFormer, and ASAP develop more efficient non-causal transport normalizers [7, 8, 9]. Sparse Sinkhorn Attention uses a Sinkhorn matrix as a block-permutation or sorting object for sparse attention, not as the token-to-token attention matrix normalized here [3]. Applying full-support column normalization directly to a causally masked token matrix produces the leaky operator in eq. (2) and fails the future-gradient probe.

The novelty claim is precise: *prefix-mass column balancing of the causal token-to-token attention matrix, with a zero-future-gradient guarantee and streaming prefix state*. The theory, scale campaign, RAPS extension, causal reservoir, role-complete head architecture, and depth-routing law developed here are specific to that operator. The relevant design space has four distinct quadrants: bidirectional token balancing, causal block sorting, ordinary causal token attention, and causal token balancing. Prefix–Sinkhorn occupies the last quadrant.

The construction also has an independent program lineage. It joins the author’s work on doubly stochastic power matrices and lossless routing [12, 13, 14] with prefix-scan state-space models and constant-memory streaming [15, 16, 17]. The move that makes the operator causal—replace a global reduction by a cumulative scan—comes directly from that lineage.

9 Limitations and next experiments

The completed results settle causality, numerical stability, the scale falsifier, delayed-copy preservation, head-role assignment, and the first depth law. Four extensions are now sharply defined.

1. **Calibrated associative retrieval.** Variable-lag extrapolation is now settled. Current multi-query recall still has a low Softmax ceiling, so the next suite must reach a stable baseline ceiling and include buried-key, collision, and pointer-chase controls that remove both recency and fixed-offset shortcuts.
2. **Original-line role scaling.** The archive-corpus role architecture is now positive across widths 32–96 and depths 1–4. The remaining scale test is the original TinyStories protocol at widths 192, 384, 512, and beyond, paired against the already completed pure Prefix curve.
3. **Bounded adaptive confirmation.** Anchor-residual routing has a small single-seed validation edge while preserving the whole free head. It requires a preregistered three-seed test before promotion.
4. **Optimized streaming kernel.** The mathematical state is simple—per-key cumulative log mass for balanced heads—but the current reference implementation prioritizes auditability. A fused scan kernel should compare wall-clock decoding and memory traffic against Softmax KV-cache attention.

These are extensions of a settled core, not repairs to it. The operator is causal, novel, scale-stable, and architecturally usable. The remaining work expands the regime in which its role-complete form is best.

10 Conclusion

Causal Prefix–Sinkhorn solves a real incompatibility: token-level column balancing and autoregressive causality can coexist only when the normalizer closes at the revealed prefix. The completed theory shows exactly what one finite step does. It forces a diagonal mode, and repeated exact causal balancing has the identity as its only bistochastic endpoint. This turns an initially informal “online Sinkhorn” picture into a sharper object: a causal allocation-control primitive.

The scale experiment then removes the central empirical doubt. The gain does not decay with width; it stabilizes near 1.18 bpb through width 1024. The long-range stress tests remove the central architectural doubt. Prefix balance should not replace every attention route. Local, balanced, and free heads have different jobs, and one untouched free head is sufficient to preserve perfect lag-48 copy. Across variable lags, the balanced role-complete foundation is the strongest tested extrapolator, retaining 0.9293 mean accuracy beyond its training range and 0.9020 at lag 80. On sealed three-seed tests, role-complete and depth-routed programs improve over Softmax on every seed; at 1200 steps the validation gains rise to 0.2507 and 0.2388 bpb for the local-control and Prefix-preserving foundations, and the two-seed architecture sweep remains positive across all tested widths and depths.

The resulting architecture is simple: specialize the foundation, preserve a whole unrestricted head, and read out through free causal Softmax. Prefix–Sinkhorn is no longer merely an alternative normalizer. It is a scale-stable causal routing organ with a precise place inside a stronger attention system.

Reproducibility statement

The release accompanying this draft contains the complete Python package, operator implementations, streaming reference, corpus manifest, synthetic generators, raw JSON outputs, human-readable reports, figure data, and all tests. Run

```
cd attention-moonshot-final
PYTHONPATH=src pytest -q
```

to reproduce the 69-test acceptance suite. Headline evidence maps to `results/theory_audit/`, `results/confirmation/`, `results/route_confirmation/`, `results/route_screen/full/`, `results/variable_lag_screen/`, `results/depth_program_stage/`, and `results/depth_confirmation/`. The scale series and original TinyStories runs are preserved in the project record and v0.4 source package.

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